

Ashi Poorey

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EDUCATION

Bachelor of Science - Astrophysics

Minors - Data Science and Computational Science & Engineering

University of Illinois Urbana-Champaign

Aug 2023 - May 2027

RESEARCH

UIUC Extragalactic Astrophysics Group

Prof. K. Decker French, Margaret E. Verrico

Aug 2026 - present

- Identifying candidate changing-look AGNs from time-domain photometric and spectroscopic data, and studying their transition/recurrence cycles to characterize accretion-state variability
- Extending a sample of ZTF-monitored changing-look AGN (CL-AGN) beyond the 52-object catalog from Verrico et al. 2026, cross-matching photometric coverage against archival spectroscopic epochs to flag possible missed state transitions
- Building toward a senior thesis on this changing-look AGN project, in spring 2027

Space Astronomy Summer Program - STScI

Dr. Stephanie LaMassa, Dr. Zach Claytor

Jun 2026 - Aug 2026

- Developed machine learning models to identify rare galaxies exhibiting broad Mg II emission in Sloan Digital Sky Survey spectra, targeting signatures of fading SMBH (CL-AGN)
- Analyzed large spectroscopic datasets (~1.4 million sources) to isolate and validate a small sample of candidate objects, enabling study of black hole accretion duty cycles
- Designed scalable methods for deployment on future surveys (SDSS extensions, DESI, etc.) to discover and characterize transient active galactic nuclei in time-domain astronomy

UIS Stellar Evolution Research

Prof. John Martin, Prof. Roberta Humphreys

Feb 2026 - present

- Investigated photometric variability and evolutionary behavior of Warm Hypergiants in M31 and M33 near the Humphreys–Davidson limit and yellow void
- Analyzed time-series photometry using Lomb–Scargle periodograms and phase-folding techniques to identify periodic and quasi-periodic variability associated with stellar pulsations and mass loss
- Implementing red noise characterization methods to assess variability significance beyond false alarm probabilities, with ongoing work toward publication

UIUC Team Gaia

Prof. Jiayin Dong

Aug 2025 - May 2026

- Independently developed and implemented an exoplanet detection pipeline using ESA Gaia mission data, constructing Bayesian inference frameworks in PyTensor + PyMC
- Applied statistical methods to time-series astrometric data for exoplanet detection, with Bayesian statistics and probabilistic modeling

PUBLICATIONS / MANUSCRIPTS IN PREPARATION

Humphreys, R., et al., including **A. Poorey** "The Historic Light Curves of the Red Supergiants: Evidence for Outflows"

CONFERENCES, TALKS & PRESENTATIONS

1. **AAS Winter Meeting 2027** - Using Machine Learning to Discover Fading AGN *[poster]*
2. **APS CU*IP 2026** - Gaia Pipeline for Inferring Planetary Companions using RUWE *[poster]*
3. **STScI SASP Symposium 2026** - Using Machine Learning to Discover Fading AGN *[presentation]*
4. **CAURS 2026** - Photometric Variability of Warm Hypergiants in M31 and M33 *[poster]*
5. **UIUC URS 2026** - Gaia Pipeline for Inferring Planetary Companions using RUWE *[poster]*
6. **UIUC Astrofest 2026** - Photometric Variability of Warm Hypergiants in M31 and M33 *[poster]*

AWARDS

1. **Distinguished Undergraduate Researcher** - University of Illinois Urbana Champaign 2026
2. **Celia Mathews Elliot Prize for Writing in Science and Engineering** - Department of Physics UIUC 2025

PROGRAMS & TRAINING

STScI JWST Summer School – *High-Redshift Transients and Time Domain Astronomy* Aug 2025

- Acquired hands-on training on high-z transient science with JWST observation planning, calibration pipelines, and data analysis.
- Gained practical experience in creating JWST observing programs and processing early-universe astrophysical data using advanced infrared techniques.

OUTREACH, VOLUNTEERING & TEACHING

The Jeffries Academic Center - *Math & Physics Tutor* Jan 2025 - May 2025
Provided weekly academic support in Physics and Math courses to 8 students

UIUC College of Liberal Arts & Sciences - *Student Ambassador* Sep 2025 - present
Chosen to represent the college at outreach and media events, and assisting prospective students with academics, campus life, and opportunities in LAS

UIUC Liberal Arts & Sciences Lift Off 2025 - *Department of Astronomy booth* Aug 2025
Represented the Astronomy department, welcoming new students and answering questions about courses, research opportunities, and campus resources

UIUC ASTR 199 Fall 2026 - *Undergraduate Research Panel* Sep 2026
Served on a panel to share experiences with undergraduate research in helping new students understand the opportunities available to them and how to get started